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Prof. Guangtao Zhai
Editor-in-Chief
Displays
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Dear Professor Zhai,

We transfer the manuscript “**Pseudo-3D Heightfield Reconstruction from Endoscopic Video via Normal-Guided Multi-View Optimization and Metric Scale Gauge Injection**” to *Displays* as a Research Article. The revised manuscript treats the heightfield as a surface that can be displayed and, separately, as a surface whose measurement error is reported.

Why Displays. The useful output is a renderable mesh for intra-operative overlay and measurement, not a disparity map alone. A collapsed cone or planar slab cannot be visualized as a surgical surface and cannot be used to measure physical lengths. This addresses a fundamental display problem: visual signal processing, an un-collapsed 3D surface, and explicit metric scale recovery.

What changed for this transfer.

- **Normal-guided branch disambiguation:** The heightfield normal selects one of the two branches of a small-baseline homography. On the evaluated pairs the selected translation agrees with the reference at median $|t^\top t_{gt}| = 0.984$.
- **Multi-stride co-visibility Bundle Adjustment (BA):** We replace sequential dead-reckoning pose chaining with a multi-stride co-visibility graph across non-adjacent keyframes, optimizing SE(3) Lie-algebra camera poses and triangulated 3D points jointly under a robust Huber loss.
- **Metric scale gauge:** On the 13 real sequences with a visible 3 mm board, full-gauge median ATE is 4.49 mm (VGGT-1B 11.88 mm, ORB-SfM 41.98 mm, same 13 sequences). Marker-free BA does not beat VGGT-1B on the 17-sequence median (29.88 vs. 16.86 mm). On s10097, BA is 24.86 mm and VGGT-1B is 37.25 mm.
- **Stereo validation and baseline gauge:** On public EndoNeRF, the calibrated stereo variant achieves an overall AbsRel of 0.0285, outperforming FoundationStereo (0.0321) and Stereo SGM (0.0323). On the chessboard benchmark, sparse gauge triangulation yields a held-out median MAE of 1.45 mm, outperforming COLMAP (3.41 mm) and VGGT (4.65 mm).

Declarations. This manuscript is not under consideration elsewhere. All authors have approved the submission. The study uses public datasets and in-house bench-top phantom recordings; it does not involve new human-subject or animal experiments. The authors have no competing interests. Code, evaluation scripts, and result files are at <https://github.com/liranyang123456-commits/re3d-heightfield>. Full in-house sequences remain available from the corresponding author.

We thank you for considering this work.

Sincerely,

Ranyang Li, et al.